

3RD HARSH-ENVIRONMENT MASS SPECTROMETRY WORKSHOP

Test of the a Miniature Double-Focusing Mass Spectrometer for the Variable Specific Magnetoplasma Rocket (VASIMR) at the Advanced Space Propulsion Laboratory (ASPL)

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ABSTRACT

The test of a miniature double focusing mass spectrometer with just 8mm of radius was performed at the Advance Space Propulsion Laboratory (ASPL) from Johnson Space Center (NASA). The experiments were conducted to provide residual gas analysis capabilities and single ion monitoring to the VX-10 vacuum chamber where NASA's new plasma propulsion system: the Variable Specific Impulse Magnetoplasma Rocket (VASIMR) is being developed. The miniature mass spectrometer tested is an alpha version of the Integrated Leak Detector System (ILD 50) developed by Mass Sensors Inc. and is a smaller version of the original CDFMS prototype ($r = 20\text{mm}$) developed at the University of Minnesota . The unit demonstrated its usefulness for both residual gas analysis and single ion monitoring, achieving almost its theoretical resolving power of 40 @FWHM and designed mass range of 50 Da. The mass spectrometer was able to continue operational even when the short pulses of plasma were being generated.